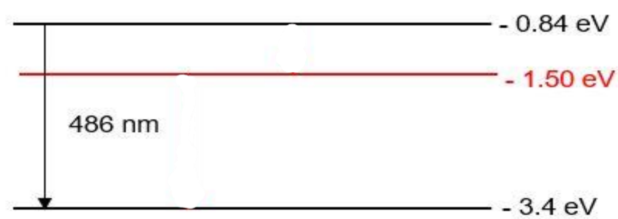
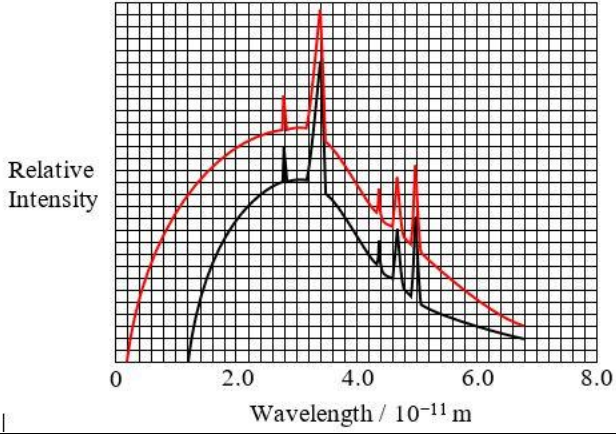
7	a	{Similarity} The discrete lines of both absorption and emission spectrum occur at same frequencies, {Difference} the absorption has dark lines against a continuous spectrum whereas the emission spectrum has coloured lines against a black background.	B1 B1
	bi	E = hf = $\frac{hc}{\lambda}$ $= \frac{(6.63\times10^{-34})\times(3.0\times10^{8})}{656\times10^{-9}\times(1.6\times10^{-19})}$ = 1.90 eV	M1 A1
	bii	 Energy level labelled -1.50 eV and nearer to -0.84 eV	A1
	c	0.92 eV Nothing happens to the lithium atom (ie. it stays in the -5.02 eV level) 0.92 – (5.02 – 4.53) = 0.43 eV Atom will be excited to the -4.53 eV level	A1 B1 A1 B1
	di	When the highly energetic electrons knock out the electrons in the inner shell of the atoms, leaving a vacancy. Electrons in the next higher energy levels transit down to the vacancy and X-ray photons are produced with energy equal to the energy difference between the 2 energy levels.	B1 B1
	dii	$eV = \frac{hc}{\lambda_o}$ $V = \frac{hc}{e\lambda_o} = \frac{(6.63 \text{ x } 10^{-34})(3.0 \text{ x } 10^{8})}{(1.6 \text{ x } 10^{-19})(1.2\times10^{-11})}$ $= 1.04\times10^{5}$ V = 104 kV	C1 A1
	diii	Same characteristic wavelengths, lower threshold wavelength, higher intensity	A1

			
8	**(a)**	There are 3 positively charged particles	A1